



Kubernetes Error 5: Evicted Pods

What is an Evicted Pod ?

- An Evicted Pod is a pod that Kubernetes forcibly removes from a node because the node doesn't have enough resources to keep it running.

Identify Evicted Pods

```
kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-abc	0/1	Evicted	0	5m



Evicted Pod Troubleshooting Flow

Pod Status = Evicted

↓
kubectl describe pod <pod-name>

↓
Check Eviction Reason

↓
MemoryPressure?

↓
DiskPressure?

↓
Ephemeral Storage Full?

↓
Check Node Resources
(kubectl top node)

↓
Free Resources or Increase Capacity

↓
Adjust Resource Requests/Limits

↓
Delete Evicted Pod

↓
New Pod Created Successfully 



Kubernetes Pod Eviction — Common Reasons & Solutions

 Cause	 Description	 Solution	 Useful Commands
 Disk Pressure	Node disk is almost full	Free disk space, remove unused images/logs, or expand the disk	kubectl describe node <node-name> df -h
 Memory Pressure	Node runs out of RAM	Free memory, optimize workloads, or add more memory/nodes	kubectl top node free -h
 Ephemeral Storage Full	/var/lib/containerd or /var/lib/kubelet is full	Clean unused images, containers, and logs or increase storage	df -h crictl images
 Node Resource Exhaustion	CPU, memory, or storage exceeds node capacity	Scale the cluster, redistribute workloads, or optimize resource usage	kubectl top node
 Resource Requests Too High	Pods request more resources than available	Reduce CPU/memory requests or add more nodes	kubectl describe pod <pod-name>
 Too Many Pods on Node	Node reaches its maximum Pod capacity	Schedule Pods on other nodes or scale the cluster	kubectl describe node <node-name>

Useful Troubleshooting Commands

 Command	 Purpose
kubectl get pods	List pods and identify evicted pods
kubectl describe pod <pod-name>	View eviction reason and events
kubectl top node	Check node CPU and memory usage
kubectl describe node <node-name>	Check node conditions (MemoryPressure, DiskPressure, PIDPressure)
kubectl top pod	Check pod resource usage
kubectl get events --sort-by=.metadata.creationTimestamp	View recent cluster events

 Command	 Purpose
<code>kubectl delete pod <pod-name></code>	Remove an evicted pod

Interview Answer

How do you troubleshoot an Evicted Pod ?

- Run **`kubectl describe pod <pod-name>`** to identify the eviction reason.
- Check node conditions using **`kubectl describe node <node-name>`**.
- Verify disk, memory, and ephemeral storage usage.
- Review **CPU** and **memory** utilization with **`kubectl top node`** and **`kubectl top pod`**.
- Clean up unused images/logs, optimize resource requests, or add more nodes if needed.

Simple Answer

-  A Pod is evicted when the kubelet removes it to protect node stability, typically due to disk pressure, memory pressure, ephemeral storage exhaustion, or overall resource exhaustion.
-  The first step is to inspect the eviction reason with **`kubectl describe pod`**, then check node conditions and resource usage before freeing resources or scaling the cluster.